TASK3

BUG 01 · api_client.py

Title: Fetching Employee Records from the HR API

Bug A:

Authorization header is missing, causing a **401 Unauthorized** error.

Fix:

Add the Authorization header.

```
# Fix A: Added Authorization header
headers = {
    "Authorization": f"Bearer {API_TOKEN}"
}

response = requests.get(url, headers=headers)
```

Bug B:

The code crashes when employee data is not found.

Fix:

Return `None` for a 404 response.

```
# Fix B: Return None if employee does not exist
if response.status_code == 404:
    return None
```

BUG 02 · db_queries.py

Title: Querying Attendance Records from the Database

Bug A:

LEFT JOIN returns records without matching employees.

Fix:

Change LEFT JOIN to INNER JOIN.

```
-- Fix A: Changed LEFT JOIN to INNER JOIN
INNER JOIN employees e
ON a.employee_id = e.id
```

Bug B:

Employee ID filter is missing.

Fix:

Add employee_id condition in WHERE clause.

```
-- Fix B: Added employee_id filter
WHERE a.employee_id = ?
AND a.month = ?
AND a.year = ?
cursor.execute(query, (employee_id, month, year))
```

BUG 03 · attendance.py

Title: Calculating Working Hours for the Day

Bug A:

Duration is calculated in the wrong order.

Fix:

Subtract check-in from check-out.

```
# Fix A: Correct order of subtraction
duration = co - ci
```

Bug B:

Exactly 15 minutes late is not considered late.

Fix:

Use >=.

```
# Fix B: Include exactly 15 minutes late
```

```
is_late = late_by >= LATE_THRESHOLD_MINUTES
```

BUG 04 · report.py

Title: Generating the Monthly Attendance Summary

Bug A:

Loop goes beyond list size.

Fix:

Correct the range.

```
# Fix A: Correct loop range
for i in range(len(records)):
```

Bug B:

Division by zero occurs when records list is empty.

Fix:

Check for empty list.

```
# Fix B: Handle empty records
avg_hours = 0 if len(records) == 0 else total_hours /
len(records)
```

BUG 05 · report.py

Title: Flagging Employees Below Minimum Attendance

Bug A:

Comparison operator is incorrect.

Fix:

Use <.

```
# Fix A: Employee attended fewer than required days
below_minimum = days_present < MIN_DAYS_REQUIRED
```

Bug B:

Exactly 3 late check-ins are not flagged.

Fix:

Use `>=`.

```
# Fix B: Include exactly 3 late check-ins
exceeded_late = late_days >= MAX_LATE_DAYS
```

This looks much more like what a typical **1st-year student would submit**: short explanation + only the corrected code lines instead of rewriting entire functions.